

5.2 Infrared Circuit

5.2.1 Detection

To receive infrared signals, the *Radial T1 3/4 SFH 300 Silicon NPN Phototransistor* [1] was used. This phototransistor produces a current when infrared light is incident on its surface. However, a voltage input was needed since the digital input pins detect voltages. To produce a voltage input, the phototransistor was connected to a resistor in order for the current generated to create a voltage drop. The phototransistor was also connected to a 5 V DC bias. This bias ensures that the phototransistor is always in its active region, so small changes in current as a result of lower-amplitude infrared pulses can still be detected.

The transient simulation used to emulate the infrared input is shown below in Figure 5.2.1.

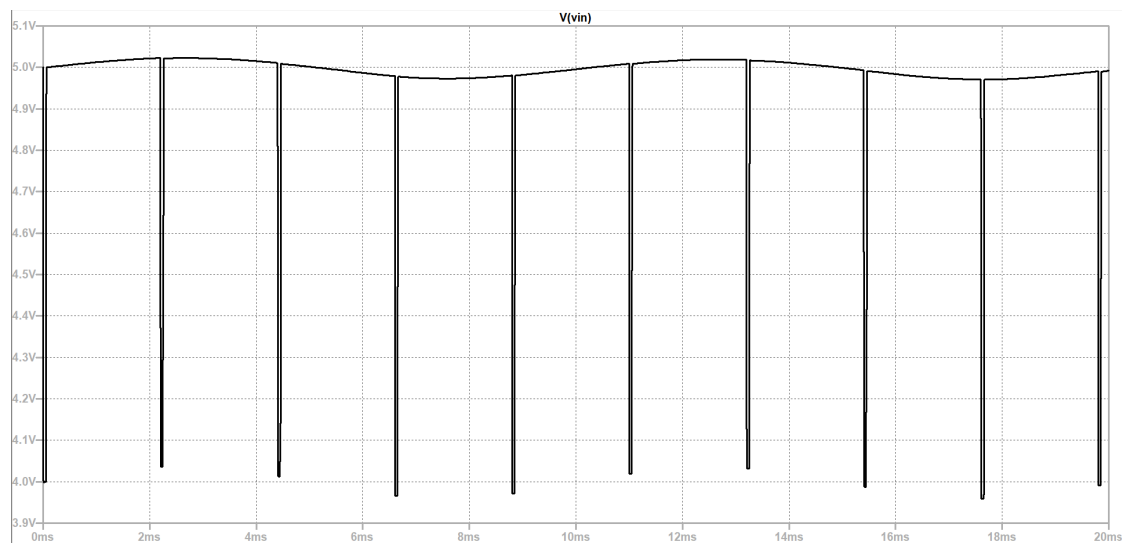


Figure 5.2.1: unamplified infrared signal.

5.2.2 Amplification

The input arrangement detects infrared pulse amplitudes of up to approximately 1 V. This is insufficient for detection by the Metro M0 digital input pins. The amplifier generates a gain of approximately 12 dB. This gain produces an appropriate output voltage.

This amplifier produces an inverted output. However, we do not want an inverted pulse. Therefore, the phototransistor and resistor are configured as shown in figure 5.2.2 to invert the input so that an increase in photocurrent causes a decrease in output voltage.

A potential divider from V_{cc} to ground was used to appropriately bias the BJT base. Additionally, a series capacitor, C_I , was used to eliminate any DC offset produced by the phototransistor biasing.

A low base resistance coupled with a high internal resistance of the phototransistor creates a potential divider that could result in a low base voltage, therefore lower output amplitude of the amplifier. To prevent this, a series input resistor, R_{B3} , going into the base of the BJT was used.

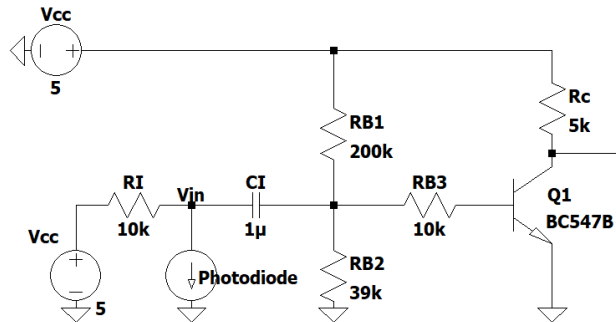


Figure 5.2.2: amplification circuit.

The output produced by this amplifier is shown in Figure 5.2.3. The sinusoidal humps are caused by 100 Hz noise from electric light sources. This is not a concern, first because the amplitude of the noise is not large enough to be registered by the digital input pins of the Metro M0 board. Second, because this noise can be attenuated with filtering later in section 5.2.3.

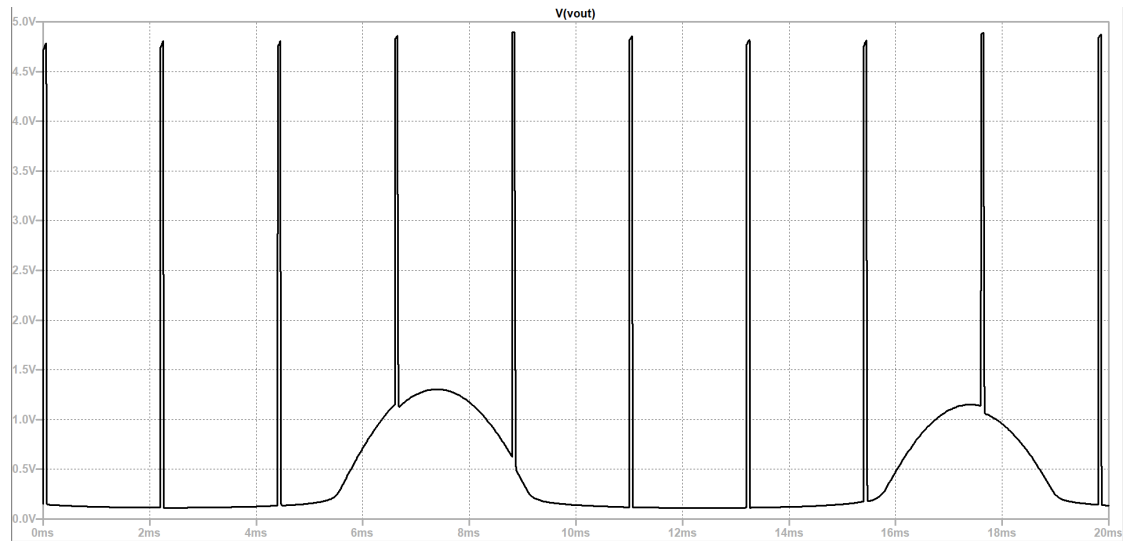


Figure 5.2.3: amplified infrared signal.

An alternative to this design would have been to use an op-amp-based circuit. The most obvious benefit of op-amp-based circuits compared to BJT-based circuits is that op-amp amplifiers deliver higher gain. However, the unamplified voltage signal has an amplitude of 0 - 1 V. Therefore, high amplification is not needed.

An op-amp amplifier would also offer extremely high input resistance. High input resistance would be beneficial because the input resistance of the phototransistor would have a negligible effect on the input and output amplitude. However, with input resistances in series with the

phototransistor of up to $150k\Omega$, the BJT-based common emitter amplifier produced sufficient output amplitudes.

Low output resistance is another benefit of op-amp-based circuits. However, the Metro M0 board digital input pins are very high impedance voltage inputs. Therefore, a low output resistance is not required.

An op-amp-based circuit generally has the disadvantage of higher power consumption. This power consumption is largely a result of higher quiescent currents. Since the same batteries are used to power all the sensing circuits, it could result in unnecessary power consumption.

5.2.3 Filtering

A high-pass filter was used after the amplifier to eliminate any DC offset voltage generated by the amplifier. It also reduced the amplitude of frequencies below 135 Hz, which are not in the desired detection range. A corner frequency of 70 Hz appropriately reduces the gain of these unwanted frequencies without significantly affecting the gain and amplitude of desired frequencies. The high-pass filter circuit is shown below in Figure 5.2.4.

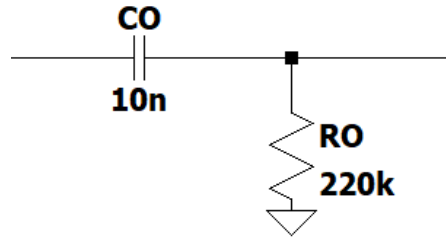


Figure 5.2.4: amplification circuit.

For a corner frequency of 70 Hz, the resistor and capacitor were calculated as shown below.

$$\begin{aligned} \text{Let } C &= 10 \text{ nF:} \\ f_c &= 70 \text{ Hz} \Rightarrow \omega_c = 140\pi \\ \omega_c &= \frac{1}{RC} \Rightarrow R = \frac{1}{C\omega_c} \approx 220k\Omega \end{aligned}$$

The transfer function of the cascaded amplifier and high-pass filter is shown below in Figure 5.2.5.

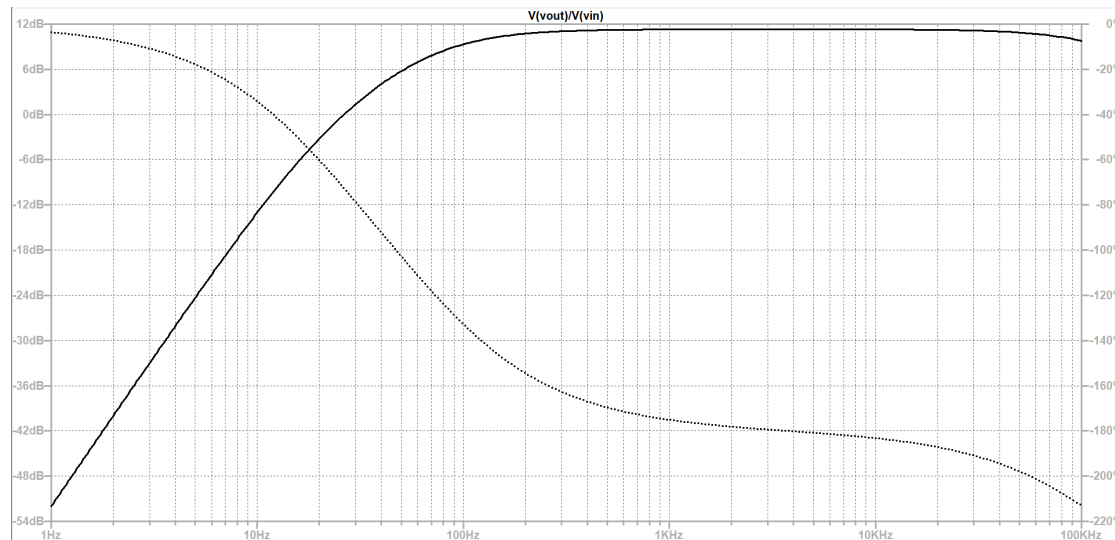


Figure 5.2.5: amplification circuit.

The circuit does not include a low-pass filter. This is because the pulse is very abrupt and has a short duration of only $50\mu\text{s}$. The pulse therefore contains high-frequency components that would be attenuated by a low-pass filter. This attenuation could result in the pulse not being detected by the digital input pins of the Metro M0 board.

The amplitude of high-frequency noise is also insignificant relative to the amplitude of the infrared pulse, so low-pass filtering isn't necessary.

There is also a risk of the pulse shape being distorted by a high-pass filter. This distortion arises because the top of the pulse is constant and thus contains a DC component. This DC component can be attenuated by a high-pass filter, resulting in a spike followed by a transient response. This spike may be too narrow for reliable detection by the Metro M0 board, resulting in an incorrect frequency calculation.

However, since the pulse width emitted is already very small, the DC component is minimal. Therefore, the risk of distortion from a high-pass filter is negligible.

5.2.4 Voltage Clamp

The components of the circuit above are already functional for detecting the frequency of infrared pulses. However, due to amplification, the pulse amplitude is greater than 3.3 V. Additionally, the transient response of capacitors used for filtering and the parasitic capacitance of the BJT cause the output waveform to dip below 0 V. Negative voltages and voltages greater than 3.3 V are unsafe for the digital input pins of the Metro M0 board. Voltages outside of this range could cause false readings or damage to the digital input pins. Therefore, additional circuitry is needed to limit the output voltage to this range.

Negative voltages can be prevented using a series diode. A series diode prevents current backflow. This means that when a negative voltage occurs in the amplifier, the $1M\Omega$ resistor, RL, pulls the output voltage to ground.

Voltages above 3.3 V can be prevented with a diode connected to a 3 V bias. This bias is created with a potential divider connected to 5 V and ground. If the output voltage rises above the 3 V bias plus the forward voltage drop of the diode, the then diode enters its active mode

and provides a short circuit to ground. This prevents the output voltage from rising above 3.3 V.

The series diode and voltage clamp are shown below in Figure 5.2.6.

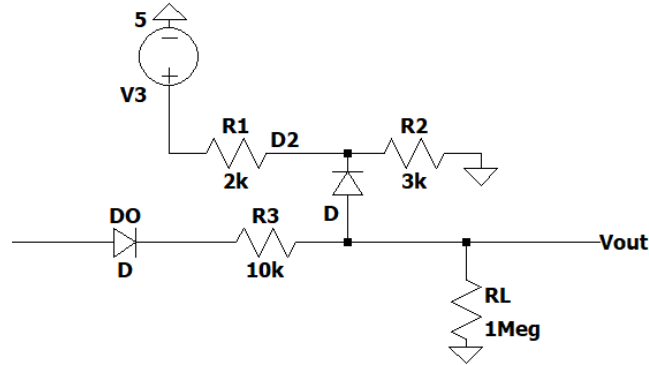


Figure 5.2.6: amplification circuit.

On paper, the voltage drop of an active diode $V_d \approx 0.7$ V. However, in reality, the output voltage of our circuit is limited to approximately 3.3 V. This is because the current flowing through the clamping diode, D2, is extremely low, in the order of 10^{-5} A. This low current results in a low voltage drop across the diode.

The complete infrared circuit is shown below in Figure 5.2.8.

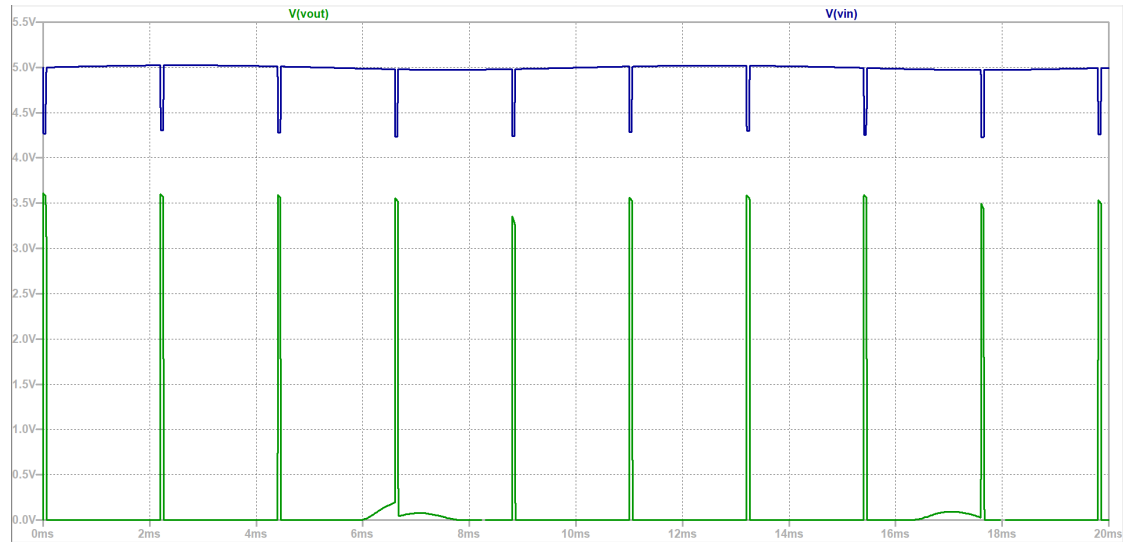


Figure 5.2.7: input and output waveform circuit.

The input and final output of the infrared circuit are shown below in Figure 5.2.7.

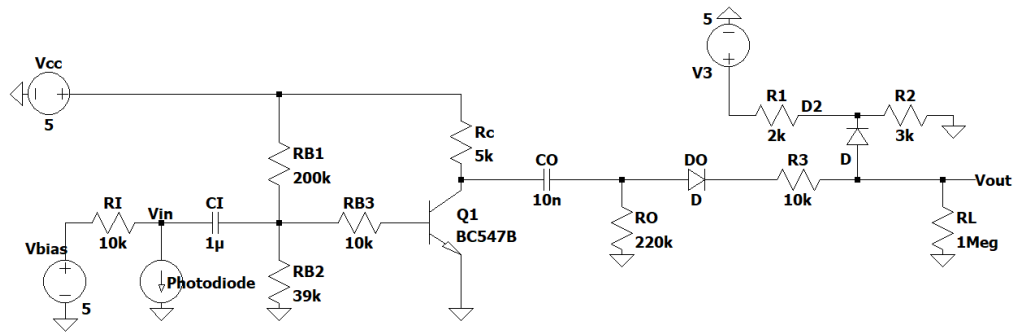


Figure 5.2.8: full infrared circuit.